Department of Software Engineering
Government College University, Faisalabad

COURSE FOLDER/FILE CONTENTS

Name of Staff Member:	DR. Muhammad Awais
Designation:	Associate Professor/Chairperson
Subject:	Programming Fundamentals
Class:	BS (SE)
Section:	N/A
Session:	2025-2029
Semester:	1st

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a. Course Objectives

To comprehend the basic programming logic and basic programming concepts. To provide a launching pad for students for advanced programming languages and logic subjects. Further, it is up to the university to choose any language for the practical/Lab purpose but that must be latest and market oriented.

CLO No.	Course Learning Outcomes	Bloom Taxonomy	Program Learning Outcomes
CLO-1	Understand basic problem solving steps and logic constructs	C2 (Understand)	PLO-3 (Problem Analysis)
CLO-2	Apply basic programing concepts	C3 (Apply)	PLO-5 (Modern Tool Usage)
CLO-3	Design and implement algorithms to solve real world problems	C3 (Solve)	PLO-4 (Design/Development of Solutions)

b. Course Contents

Introduction to problem-solving, Introduction to programming, the role of compiler and linker, introduction to algorithms, basic data types and variables, input/output constructs, arithmetic, comparison and logical operators, conditional statements and execution flow for conditional statements, repetitive statements, and execution flow for repetitive statements, lists and their memory organization, introduction to modular programming, function definition and calling, Arrays, Pointers, string and string operations, pointers/references, static and dynamic memory allocation, File I/O operations.

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c. Weekly plan of contents of lectures delivered

	Department of Software Engineering **Government College University, Faisalabad**			
Lecture Plan				
Course Title	Programming Fundamental			
Course Code	CCC-301		**Credit Hours**	4 (3 – 1)
Class	1st Semester		**Semester**	
Week	**Lecture**	**Topic**	**Source Book / Notes**	**Lab /Tutorials**
1	1.1 1.2 1.3	<u>Introduction to problem solving</u> Program, Algorithm & Pseudo code, Flowchart, Program development process, Programming languages, Types of codes, Language processors, Programming Techniques	Chapter#1 Deitel & Deitel C++ How to Program, Object oriented Programming using C++, Object oriented Programming in C++ by Rober Lafore.	Lab +assignment
2	2.1 2.2 2.3	<u>Introduction to C++</u> History of C++, Basic Structure of C++ Program, Header Files, Main function, Debugging, Linking, Compiling, Executing a Program, Types of error	Chapter#2 Deitel & Deitel C++ How to Program, Object oriented Programming using c++, Object oriented Programming in C++ by Rober Lafore, (Chapter#2).	Lab
3	3.1 3.2 3.3	<u>C++ Programming Basics</u> Identifier, Keywords, Data Types, Integer Overflow and Underflow, Variables, Constants, Operators, Type Casting	Chapter#3 Deitel & Deitel C++ How to Program, Object oriented Programming using c++, Object oriented Programming in C++ by Rober Lafore, (Chapter#2).	Lab +assignment
4	4.1 4.2 4.3	<u>Inputs and Output</u> Input and Output, Standard Output, Standard Input, C++ Manipulators, Precedence, const Qualifier, etc.	Deitel & Deitel C++ How to Program (Chapter#2), Object oriented Programming using c++ Chapter#4, Object oriented Programming in C++ by Rober Lafore, (Chapter#2).	Lab

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5	5.1 5.2 5.3	<u>Conditional Structures</u> Control Structures, Relational Operators, if else statement, Multiple if else if Structure, Nested if Structure, Compound Conditions, Switch Structure, Conditional Operator, goto statement.	Deitel & Deitel C++ How to Program (Chapter#2), Object oriented Programming using c++ Chapter#5, Object oriented Programming in C++ by Rober Lafore, (Chapter#3).	Lab +assignment
6	6.1 6.2 6.3	<u>Looping Structures</u> Loops, while loop, do while loops, for loops, Nested loops, continue / break statement	Deitel & Deitel C++ How to Program (Chapter#2), Object oriented Programming using c++ Chapter#5, Object oriented Programming in C++ by Rober Lafore, (Chapter#3).	Lab
7	7.1 7.2 7.3	<u>Arrays</u> Advantages of Array, Declaring Array, Initialization, Searching in Arrays, Sorting Array, Two Dimensional Array, Multidimensional Array, Enumerators, Memory representation of Arrays, etc.	Deitel & Deitel C++ How to Program (Chapter#4), Object oriented Programming using C++ (Chapter#7), Object oriented Programming in C++ by Rober Lafore, (Chapter#3).	Lab +assignment
8	8.1 8.2 8.3	<u>Functions</u> Types of Functions in C++, User Defined Functions, Passing Parameters to Functions, Local Variables, Global variables, Static Variables, Register Variable, inline function, Default argument, Array as function arguments.	Deitel & Deitel C++ How to Program (Chapter#3), Object oriented Programming using C++ (Chapter#9), Object oriented Programming in C++ by Rober Lafore, (Chapter#5).	Lab
8th week Mid term				
9	9.1 9.2 9.3	<u>Pointers</u> Addresses and Pointers, Address-of Operator &, Pointer Variables, Pointer declaration, Dereference operator, pointer initialization, pointer addition, pointer subtraction, Array of pointers	Deitel & Deitel C++ How to Program (Chapter#5), Object oriented Programming using C++ (Chapter#11), Object oriented Programming in C++ by Rober Lafore, (Chapter#10).	Lab +assignment
10	10.1 10.2 10.3	<u>Pointers</u> Accessing the Variable Pointed To, Pointers and Functions, Passing Simple Variables, Passing Arrays, new & delete operator for dynamic memory allocation, Heap & Stack Memory.	Deitel & Deitel C++ How to Program (Chapter#5), Object oriented Programming using C++ (Chapter#11), Object oriented Programming in C++ by Rober Lafore, (Chapter#10).	Lab
11	11.1 11.2 11.3	<u>Built in Function</u> Conio.h Header file function, stdio.h header file functions, Math Functions etc. C++ program consisting of Multiple files, including header file (.h) and .cpp file containing main function	Object oriented Programming using C++ (Chapter#10),	Lab +assignment
12	12.1 12.2 12.3	<u>String Handling</u> Memchr(), memcmp(), memcpy(), memmove(), memset(), strcat(), strncat(), strchr(), strcmp(), strncmp(), strcoll(), strcpy(),strlen(),strpbrk(), strkok(), etc.	Deitel & Deitel C++ How to Program (Chapter#5), Object oriented Programming using C++ (Chapter#12), Object oriented Programming in C++ by Rober Lafore, (Chapter#7).	Lab
13	13.1 13.2 13.3	<u>Structures</u> Declaring a Structure, Defining Structure Variable, Accessing Members of Structure, Initialization, Array as Members of Structure	Object oriented Programming using C++ (Chapter#8),	Lab +assignment
14	14.1 14.2 14.3	<u>Structures</u> Array of Structures, Nested Structure, Union, Enumerations. <u>Templates</u> Function Templates: declaration & usage	Object oriented Programming using C++ (Chapter#8), Object oriented Programming in C++ by Rober Lafore, (Chapter # 14)	Lab

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		Function Templates with Multiple Arguments		
15	15.1 15.2 15.3	<u>Graphic Programming</u> Preparing for Graphics Functions Errors Getting System Information Line Width and Style Circle Drawing, etc.		Lab +assignment
16	16.1 16.2 16.3	<u>File Handling</u> File access methods, Sequential and Random Access, Types of Streams, reading / writing files in character, binary, multiple lines in sequential fashion. File reading and writing using seekg() and seekp() functions	Deitel & Deitel C++ How to Program (Chapter#14), Object oriented Programming using C++ (Chapter#18), Object oriented Programming in C++ by Rober Lafore, (Chapter#12).	Lab
16th week FINAL SEMESTER EXAMINATION				

d. Attendance Record

(From Portal)

e. Copy of lecture notes

Reference books, provision in case No Reference book is available

f. Quizzes Marks and Assignment Marks

g. List of Reference Material/Books Pages/Codes

1. Starting out with Programming Logic & Degins, 4th Edition, Tony Gaddis,
2. The C Programming Language, 2nd Edition by Brian W. Kernighan, Dennis M. Ritchie
3. Object Oriented Programming in C++ by Robert Lafore
4. C How to Program, 7th Edition by Paul Deitel & Harvey Deitel
5. Problem Solving and Program Design in C++, 7th Edition by Jeri R. Hanly & Elliot B. Koffman
6. Object Oriented Programming Using C++, 2nd Edition by Tasleem Mustafa & Ahsan Raza Sattar
7. Copy of questions of assignments, quizzes, midterms and final examinations + Model solutions of all assessments tests given.

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8. Three sample graded assignments, quizzes, midterms and final examination securing max, min and average marks + Marks distribution and Grading Model
9. Complete result of the course

Miscellaneous/Optional

- Outcomes Assessment
- Detail of technology involved
- Design skills/techniques practiced
- Complete analysis of effectiveness of course and level of silks ensured in:
- Technology
- Emerging Development Paradigms
- Pertaining to Industry
- Modeling and Design

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